Testing the Fluorescence Advantage: Solar-Induced Fluorescence as an Early Indicator of Agricultural Drought Stress in Marathwada, Maharashtra

Abstract

Drought detection frameworks used for official drought declaration and agricultural insurance assessment in India rely substantially on rainfall-deficit records and the Normalized Difference Vegetation Index (NDVI), both of which respond to crop stress only after visible physiological degradation has already occurred. This study examines whether Solar-Induced Fluorescence (SIF), a satellite-derived proxy for photosynthetic activity, registers vegetation stress measurably earlier than NDVI. Using GOSIF v2 fluorescence data and cloud-screened MODIS NDVI over the Marathwada region of Maharashtra across eight growing seasons (2015–2023, excluding 2021 for data availability reasons), a quantitative lag was calculated between the two indices' post-peak seasonal decline. Rainfall-anomaly validation against a twenty-year (2001–2020) climatological baseline identifies only two of these eight years — 2015 (−21.5%) and 2018 (−18.3%) — as meeting this study's own drought threshold; the remaining six fall within roughly one standard deviation of normal. Under the threshold-crossing lag method, SIF's decline preceded NDVI's decline in seven of the eight years studied; the eighth (2018) showed an essentially flat, slightly negative mean lag (−1.1 days), a genuine exception rather than a rounding artifact. The hypothesis that drought conditions amplify this lag (H3) is again not supported at the larger sample size: the two drought years averaged a smaller lag (7.6 days) than the six normal years (15.0 days). An independent, methodologically distinct cross-correlation lag estimate corroborates the qualitative direction of this finding without reproducing its exact numbers: the correlation-maximizing lag was non-negative in all eight years (never favoring NDVI leading SIF) and strictly positive in four of eight, with the other four — including 2018 — best aligned at zero lag. A case-resampling bootstrap (2,000 replicates per year) confirms this weaker but still real claim precisely: the estimated lag was at or above zero in 100% of replicates in every single year, while the specific day-count magnitude carries wide uncertainty, and every pairwise between-year comparison is statistically indistinguishable. District-level spatial analysis finds mean SIF and rainfall anomaly significantly correlated across the 64 district-year observations (8 districts × 8 years): Pearson $r = 0.567$, Spearman $\rho = 0.551$, both $p < 0.0001$ — a real, still-significant relationship, but substantially weaker than the $r = 0.837$ reported from the original 24-point, 3-year sample, consistent with that earlier correlation having been inflated by a small, drought-heavy sample. Moran's I confirms that both variables carry genuine spatial structure, though less consistently for SIF (significant in 4 of 8 years) than for rainfall (significant in all 8 years), and the effective independent sample size behind the district correlation remains closer to the number of years than the number of district-year rows. A comparison against the one well-documented official drought-declaration date available (Maharashtra, 31 October 2018) found a notable reversal from the original 3-year study once the underlying

district boundary was corrected: NDVI now crosses its 90% decline threshold roughly five days before SIF does in 2018 specifically (not after, as originally reported), though both indicators still lead the official declaration by seven to eight weeks. This reversal, along with the near-zero 2018 lag under both lag methods, is reported directly as a genuine finding of the expanded sample rather than smoothed over.

Keywords: solar-induced fluorescence, remote sensing, drought monitoring, NDVI, agricultural stress, Marathwada

1. Introduction

1.1 Motivation

The Normalized Difference Vegetation Index is a reflectance-based measure: it quantifies how much visible and near-infrared light a plant canopy reflects, and this reflectance changes only once a plant's internal structure has already begun to visibly degrade — thinning leaves, reduced chlorophyll content, canopy stress. Solar-Induced Fluorescence is grounded in a more direct physical process. When a leaf absorbs sunlight for photosynthesis, a small fraction of the absorbed energy is not converted into chemical energy but is instead re-emitted as fluorescence, governed by the quantum yield of photosynthesis. Under physiological stress, photosynthetic efficiency declines before outward greenness does, meaning SIF is expected to register stress earlier than NDVI. Where NDVI reflects a plant's outward appearance, SIF reflects what is occurring physiologically within it.

India's drought declaration process, and the crop-insurance payout mechanism under the Pradhan Mantri Fasal Bima Yojana (PMFBY) that depends on it, relies substantially on rainfall-deficit and NDVI-based assessment. Both indicators are known to register drought only after visible crop stress has already set in, and this delay translates directly into delayed insurance payouts at precisely the point when affected farmers require them most urgently. This study treats the existence and size of a SIF–NDVI lag as an empirical, testable question, with the aim of establishing whether such a lag constitutes a physics-based case for an earlier-warning indicator than those presently used in policy.

1.2 Physical Basis of Solar-Induced Fluorescence

When a leaf absorbs photosynthetically active radiation, the absorbed energy is partitioned among three competing, mutually exclusive pathways at the chlorophyll level: photochemistry (Φ_P), non-photochemical quenching or heat dissipation (Φ_{NPQ}), and chlorophyll fluorescence (Φ_F). These three pathways account for the entirety of absorbed energy, described by the conservation relationship:

$$\Phi_P + \Phi_{NPQ} + \Phi_F = 1$$

Under well-watered, physiologically normal conditions, a large share of absorbed energy is directed toward photochemistry, with a comparatively small and relatively stable fraction re-emitted as fluorescence. Under drought or heat stress, the photochemical pathway is down-regulated as the photosynthetic apparatus protects itself from damage, while the non-photochemical quenching pathway is up-regulated to dissipate the resulting surplus energy as heat. This re-partitioning alters the fluorescence yield in a manner that is measurable before any outward change in canopy structure or greenness occurs, which is the physical basis for SIF's expected earlier-warning capacity relative to reflectance-based indices such as NDVI (Figure 1).

The magnitude of the fluorescence signal actually detected by a satellite sensor, denoted SIF, is the product of three quantities:

SIF = APAR × Φ_F × f_esc

where APAR is the absorbed photosynthetically active radiation, Φ_F is the fluorescence quantum yield described above, and f_esc is the fraction of emitted fluorescence photons that escape the canopy without being reabsorbed. Because chlorophyll fluorescence is several orders of magnitude weaker than reflected sunlight at the same wavelengths, it cannot be measured through simple radiance detection. Satellite-based SIF retrievals, as performed by instruments such as GOME-2, OCO-2, and TROPOMI, instead exploit narrow, naturally dark absorption features in the solar spectrum — Fraunhofer lines — where the additional radiance contributed by fluorescence emission can be isolated from the reflected background. The GOSIF product used in this study is not itself a direct satellite retrieval, but a statistically modeled reconstruction combining discrete OCO-2 SIF soundings with continuous MODIS reflectance data and meteorological reanalysis to produce spatially and temporally continuous global coverage.

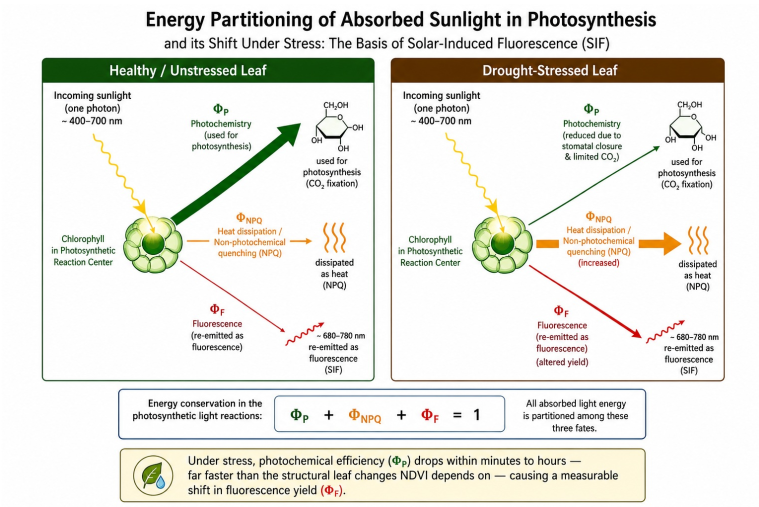

Figure 1

Figure 1. Schematic representation of leaf-level light energy partitioning among photochemistry, non-photochemical quenching (heat dissipation), and chlorophyll fluorescence, under well-watered versus drought-stressed conditions. AI was used to help generate this image, but the concept and every detail in it are mine.

1.3 Physical Basis of NDVI

The Normalized Difference Vegetation Index is computed from surface reflectance in the red and near-infrared bands:

NDVI = (NIR − RED) / (NIR + RED)

This formulation exploits a specific optical property of healthy plant tissue. In the red portion of the spectrum, chlorophyll pigments within the leaf's palisade mesophyll strongly absorb incoming light to drive photosynthesis, resulting in low red reflectance. In the near-infrared, by contrast, the internal leaf structure — specifically the irregular air-cell interfaces within the spongy mesophyll layer — causes strong multiple scattering due to the refractive-index mismatch between cell walls and intercellular air spaces, resulting in high near-infrared reflectance that is largely independent of chlorophyll content (Figure 2). A canopy with abundant, healthy, chlorophyll-rich foliage therefore produces a large contrast between low red and high near-infrared reflectance, yielding a high NDVI value; a canopy undergoing structural degradation — thinning leaves, reduced leaf area, senescence — reduces this contrast and lowers NDVI.

Because this signal depends on the plant's outward structural and pigment state, NDVI necessarily lags behind the internal physiological changes that precede visible degradation. Cloud and cloud-shadow contamination, arising from Mie scattering, further complicates the NDVI signal in the optical domain, an issue explicitly addressed in this study via MOD13Q1's `SummaryQA` per-pixel quality flag prior to any analysis (Section 3.1). This structural, appearance-based basis for NDVI stands in direct physical contrast to the internal-process basis of SIF described in Section 1.2, and this contrast is the central physical premise investigated throughout this study.

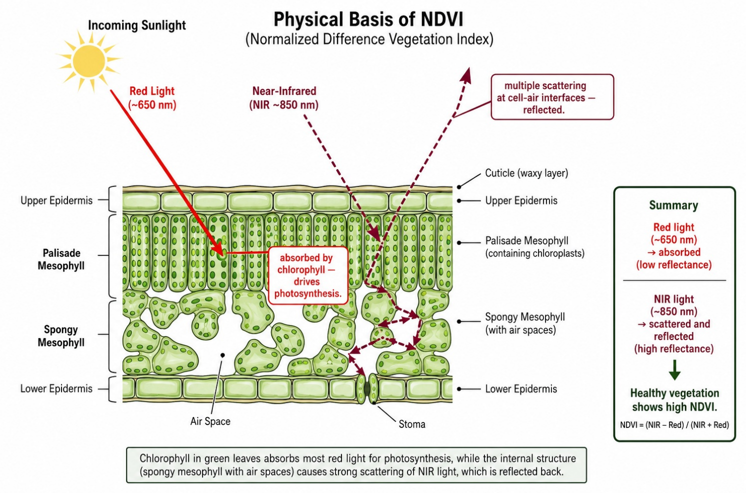

Figure 2

Figure 2. Leaf cross-section illustrating red-light absorption by chlorophyll in the palisade mesophyll versus near-infrared scattering at spongy mesophyll air-cell interfaces, the physical basis of the reflectance contrast measured by NDVI. AI was used to help generate this image, but the concept and every detail in it are mine.

1.4 Research Questions

RQ1: Does SIF decline measurably earlier than NDVI during documented drought-onset periods in the study region?

RQ2: Is the SIF–NDVI lag, where present, consistent across the study region, or does it vary meaningfully by geography?

RQ3: How does the timing of SIF-based stress onset relate to independently measured rainfall deficit across drought and normal years?

1.5 Hypotheses

H1: SIF declines significantly before NDVI during a given drought episode.

H2: The magnitude of SIF–NDVI divergence is not spatially uniform across the study region.

H3: Drought severity, measured independently through rainfall deficit, amplifies the magnitude of the SIF–NDVI lag.

2. Study Area

The study region comprises the eight districts constituting Marathwada, Maharashtra — Chhatrapati Sambhajinagar (Aurangabad), Jalna, Parbhani, Hingoli, Nanded, Beed (Bid in the FAO GAUL administrative dataset used for boundary definition — see Section 3.2), Latur, and Dharashiv (Osmanabad) — selected for its recurring, well-documented history of agricultural drought.

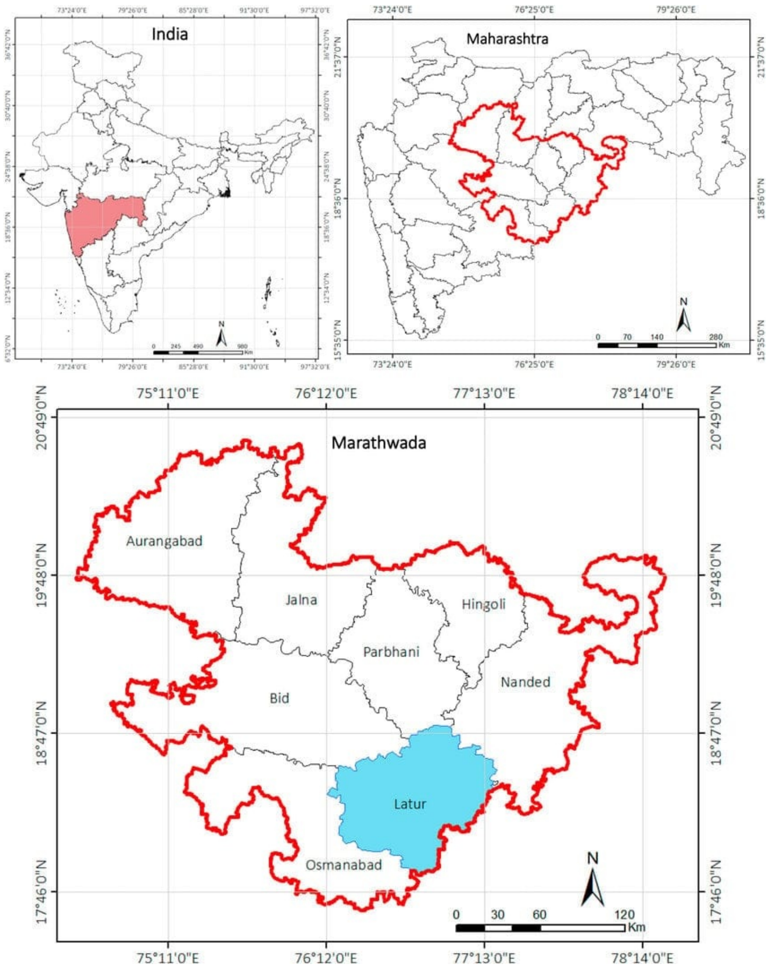

Figure 3

Figure 3. Location of the study area. (a) Maharashtra state within India. (b) Marathwada division within Maharashtra. (c) The eight constituent districts of Marathwada, with Latur district highlighted for reference.

3. Data and Methodology

3.1 Data Sources

Four datasets were used, all covering the June 1 – December 27 window (day-of-year 153–361) for eight study years — 2015, 2016, 2017, 2018, 2019, 2020, 2022, and 2023 (2021 was not included in this expansion pass):

- **SIF:** GOSIF v2, a global 0.05°, 8-day solar-induced fluorescence product (Li & Xiao, 2019), distributed by the Global Ecology Group, University of New Hampshire.
- **NDVI:** MODIS MOD13Q1, a 16-day, 250 m Vegetation Index product with an internal Maximum-Value-Composite cloud-suppression algorithm, accessed via Google Earth Engine.
- **Land cover:** MODIS MCD12Q1 (IGBP classification), used to derive a cropland mask (classes 12, 14).
- **Precipitation:** CHIRPS daily rainfall (UCSB-CHG), accessed via Google Earth Engine, for all eight study years and for a twenty-year (2001–2020) climatological baseline.

The study originally covered three years (2015, 2018, 2020); five additional years (2016, 2017, 2019, 2022, 2023) were added in a later expansion pass specifically to address the small-sample limitation noted in every earlier version of this paper's Section 6. The acquisition and processing pipeline is otherwise unchanged between the original three years and the five added years — the same GOSIF product, the same MOD13Q1/SummaryQA/cropland-mask NDVI pipeline, and the same FAO GAUL district boundary (Section 3.2) were used throughout, so the eight years are directly comparable rather than drawn from two different methodologies.

3.2 Boundary Definition and Verification

The study region boundary was defined using the FAO GAUL 2015 level-2 administrative dataset, filtered to the eight Marathwada districts and merged into a single precise polygon. This replaced an initial rectangular bounding-box definition, which was found on visual inspection to extend beyond Marathwada into neighboring Telangana districts and into Solapur and Yavatmal — areas outside the study region with distinct rainfall regimes. Both the SIF raster-clipping pipeline (using polygon-based masking) and the NDVI extraction pipeline (using the same polygon as the reducer geometry) were built against this corrected boundary.

A second, more subtle boundary error was found and fixed during the sample-size expansion described in Section 3.1: the district name list used to build this polygon from FAO GAUL's `ADM2_NAME` field used the common English spelling "Beed," which does not match FAO GAUL 2015's own spelling for this district, "Bid." Because a filter on a name that matches nothing simply returns no result rather than an error, this silently dropped Bid district from the merged region boundary and every district-level export — affecting the region-wide SIF, NDVI, and rainfall statistics reported for all eight years the first time the expanded extraction was run. This was caught by counting rows in the by-district output (56 rather than the expected 64) rather than by any error message, fixed by correcting the spelling, and the full extraction was re-run and re-verified before any of the numbers in this paper were computed. All statistics in this paper reflect the corrected, eight-district boundary.

Following the original 2015 boundary correction, the masking behavior was independently verified by overlaying the true boundary polygon directly on a sample clipped SIF raster, confirming that data was retained only within the actual district outlines and excluded elsewhere.

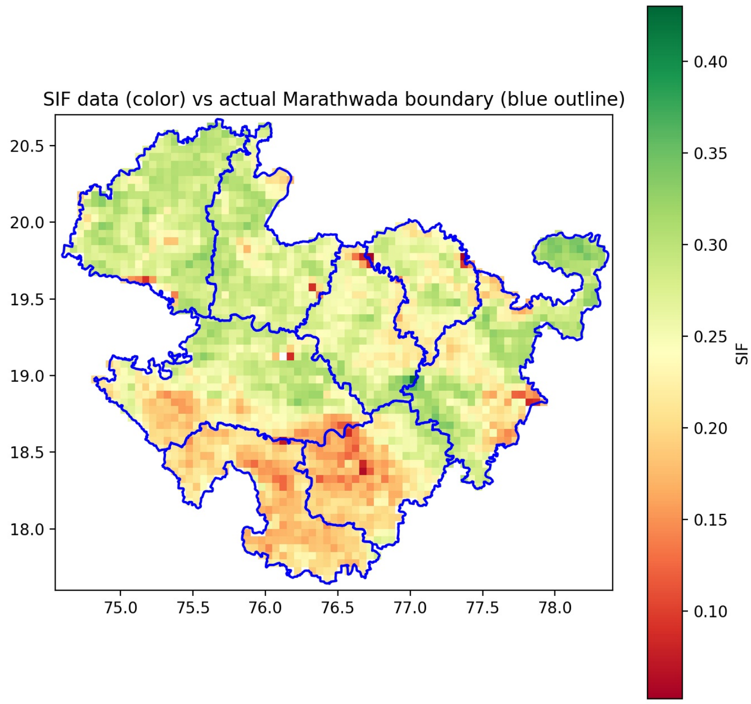

Figure 4

Figure 4. Verification of boundary masking: clipped SIF data (color) overlaid with the true Marathwada district boundary (blue outline), confirming that raster values are correctly excluded outside the study region.

3.3 Processing Pipeline

SIF GeoTIFFs were clipped to the study boundary, scaled by GOSIF's 0.0001 factor, and masked for fill-value codes (32766, water; 32767, non-vegetated/missing) prior to averaging. NDVI was extracted using MOD13Q1's `SummaryQA` band (retaining good- and marginal-quality pixels only) combined with the cropland mask. The two series, at different native temporal resolutions (8-day SIF, 16-day NDVI), were aligned using nearest-date matching within each year.

3.4 Lag Calculation

Each year's SIF and NDVI series were normalized (0–1) within-year, and the day-of-year at which each series crossed a set of decline thresholds (90%, 80%, 70%, 60%, 50% of seasonal peak) was estimated via linear interpolation between observed dates. Lag was defined as the difference between the NDVI and SIF crossing dates at each threshold.

3.5 Rainfall Validation

CHIRPS-derived seasonal rainfall totals were computed for all eight study years and compared against a twenty-year (2001–2020) climatological mean and standard deviation for the same region and seasonal window, yielding a percentage anomaly and z-score for each study year. This was extended to the district level using the same regional climatological baseline as a shared reference across all eight districts. A year is classified as a drought year in this study if its seasonal rainfall anomaly falls at least half a climatological standard deviation below the mean (anomaly z-score < -0.5) — the same implicit threshold that separated the original study's two drought years from its one normal year, now made explicit and applied uniformly across all eight years rather than assigned by external knowledge.

3.6 Cross-Correlation Robustness Check

The threshold-crossing method (Section 3.4) estimates lag from when each series crosses a given percentage of its own seasonal peak — a measure most sensitive to the *onset* of decline, particularly at high thresholds. As an independent check on whether this specific choice of method drives the reported lag values, a second, methodologically distinct lag estimate was computed using time-lagged cross-correlation, which instead asks what single time-shift best aligns the two curves' overall shape across the full post-peak decline window. For each year, both series (normalized 0–1, restricted to the post-peak decline phase from SIF's own peak onward) were interpolated onto a common daily grid, de-meaned, and cross-correlated at lags from 0 to N/4 days, where N is the length of the decline window in days — a standard bound that avoids the unreliable, boundary-pinned estimates that result from testing lags approaching the full series length. The lag maximizing the correlation between SIF(t) and NDVI(t + lag) was taken as the cross-correlation-based lag estimate for that year.

3.7 Uncertainty Quantification (Bootstrap Confidence Intervals)

The cross-correlation lag estimates in Section 3.6 are point estimates computed from a small number of discrete satellite observations per year — 14 to 18 post-peak, SIF/NDVI-matched 8-day observations, depending on the year's decline window. To quantify how much these estimates could plausibly vary given the sparseness of the underlying observation dates, each year's lag estimate was bootstrapped using case resampling: the original per-year observations were resampled with replacement 2,000 times, each replicate was re-interpolated onto a daily grid and re-processed through the identical cross-correlation procedure described in Section 3.6, and the 2.5th and 97.5th percentiles of the resulting lag distribution were taken as a 95% confidence interval around the original point estimate. An earlier version of this analysis used a moving-block bootstrap applied directly to the interpolated daily series; that approach is documented, and abandoned, in GA_Development_Log.md Entry 13, because block-shuffling a strongly non-stationary decline curve destroys the trend that produces the lag signal in the first place, collapsing every year's interval to a spurious single point at lag 0 rather than measuring genuine sampling uncertainty. The case-resampling approach used here instead resamples which underlying satellite overpass dates are represented in the fit — the more realistic source of uncertainty given an 8-day, cloud-gap-affected revisit cycle — while preserving the shape of the decline curve in every replicate.

3.8 Spatial Autocorrelation Diagnostic

The district-level Pearson/Spearman correlation reported in Section 4.5 already carries the caveat that its district-year observations are not fully independent, since the eight districts are geographically adjacent and share regional weather systems. To move beyond a qualitative caveat, Moran's I — a standard measure of spatial autocorrelation — was computed for both mean SIF and rainfall anomaly, separately for each study year, using a Queen-contiguity spatial weights matrix built directly from the FAO GAUL district polygons used throughout this study (mean of 3.25 neighboring districts per district), with significance assessed via a 9,999-permutation test.

3.9 Comparison Against the Official Drought-Declaration Timeline (RQ3)

RQ1 (Section 1.4) asks whether SIF leads NDVI; Sections 4.1–4.2 and 4.6 answer this directly. RQ3, as originally posed in this project's planning documents, asks how satellite-detected stress onset relates to independently measured rainfall deficit — a version already answered by Sections 3.5 and 4.4–4.5. A related, more policy-relevant question — how satellite-detected stress onset compares to the timing of the *official* government drought declaration that governs PMFBY payout timing — was part of this project's early research-question list (GA_Project_Report.md) but was not addressed in earlier drafts of this paper, a gap I noticed on a later pass through this project. This section closes that gap for the one study year with a single, well-documented official declaration date: the Maharashtra government's Kharif drought declaration of 31 October 2018 (day-of-year 304), covering 151 talukas across 26 districts, including all eight Marathwada study districts (Zee News, 2018; Economic and Political Weekly, 2018). A comparably well-documented, single declaration date specific to any other single study year could not be located, so this comparison remains a single-year (2018) illustration of the general gap between satellite signals and the official process, not a claim generalized across all eight years.

4. Results

4.1 Seasonal SIF–NDVI Trajectories

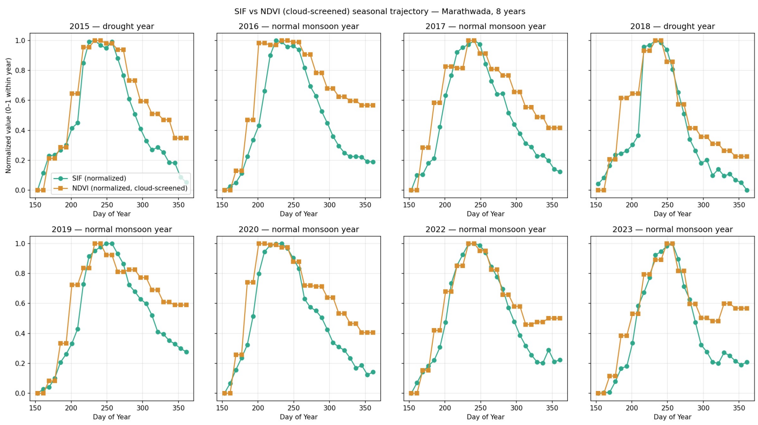

Figure 5

Figure 5. Normalized SIF and NDVI seasonal trajectories, Marathwada, all eight study years (2015–2023), labeled by drought classification (Section 3.5).

In six of the eight years, SIF began its post-peak decline visibly before NDVI, with NDVI remaining near its peak value for a longer period after SIF had already started dropping. 2018 is a clear visual exception — the two curves stay close together through the decline, tracking each other far more tightly than in any other study year, consistent with the near-zero mean lag reported for 2018 in Section 4.2. This single-year exception is discussed directly in Section 5 rather than smoothed into the general pattern.

4.2 Quantitative Lag Analysis

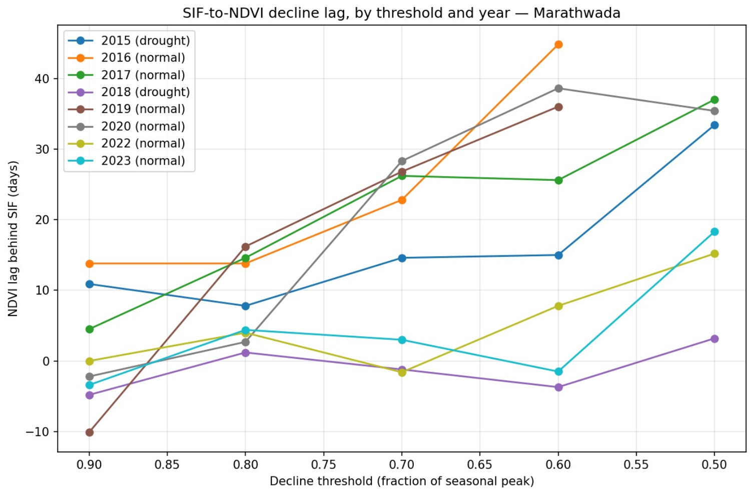

Figure 6

Figure 6. SIF-to-NDVI decline lag (days), by decline threshold and year.

Mean lag across all resolved thresholds was 16.3 days (2015), 23.8 days (2016), 21.6 days (2017), −1.1 days (2018), 17.2 days (2019), 20.6 days (2020), 5.1 days (2022), and 4.2 days (2023). Seven of the eight years show a positive mean lag (SIF leading NDVI); 2018 is the sole exception, with a small, essentially flat lag that is marginally negative rather than zero — meaning at most thresholds SIF and NDVI crossed the same decline point within a few days of one another, in either order depending on the threshold. Grouped by drought classification (Section 3.5), the two drought years averaged a smaller lag (7.6 days) than the six normal years (15.0 days) — the opposite of the direction predicted by H3, and consistent with the direction (though not the exact numbers) reported in every earlier version of this study.

4.3 Spatial Analysis — SIF

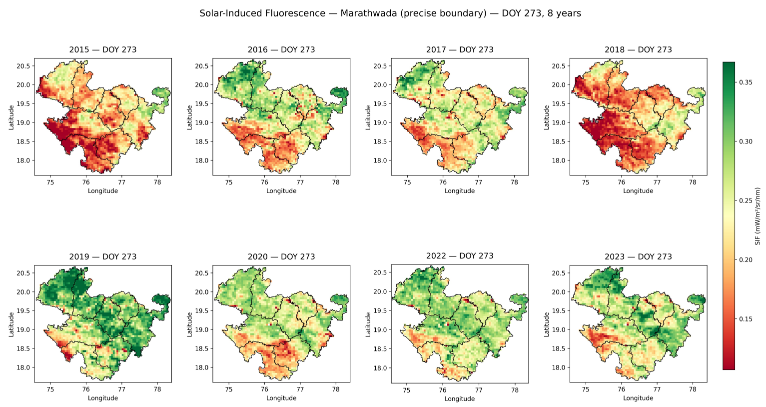

Figure 7

Figure 7. Spatial SIF distribution, day-of-year 273, all eight study years, masked to the precise Marathwada boundary.

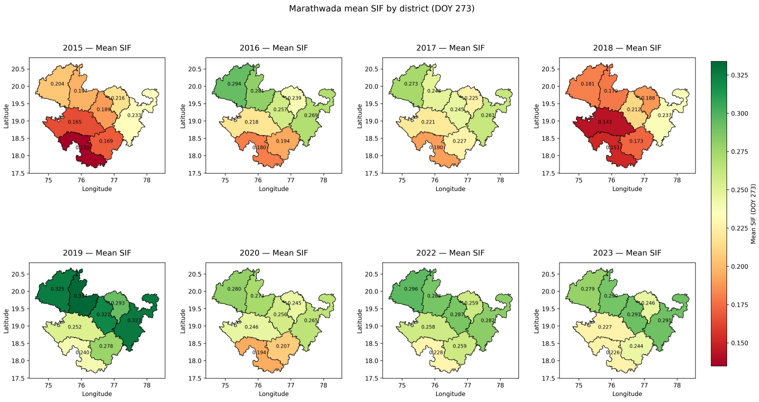

Figure 8

Figure 8. Mean SIF by district, day-of-year 273, all eight study years.

Osmanabad records the lowest district-level mean SIF in every one of the eight years studied (0.135, 0.180, 0.190, 0.151, 0.240, 0.194, 0.228, 0.226, in year order 2015–2023) — a pattern that held in all three years of the original study and now holds across the full eight-year record, strengthening it from a three-for-three to an eight-for-eight replication. The highest district-level mean SIF is more variable across the expanded sample than the original three years suggested: Nanded was highest in the original study's two drought years, and remains highest in three of the eight years overall (2015, 2018, 2023), but Aurangabad is highest in four years (2016, 2017, 2020, 2022) and Jalna in one (2019) — so "Nanded consistently highest" does not hold up as a general finding once more years are considered, even though "Osmanabad consistently lowest" does.

4.4 Rainfall Validation

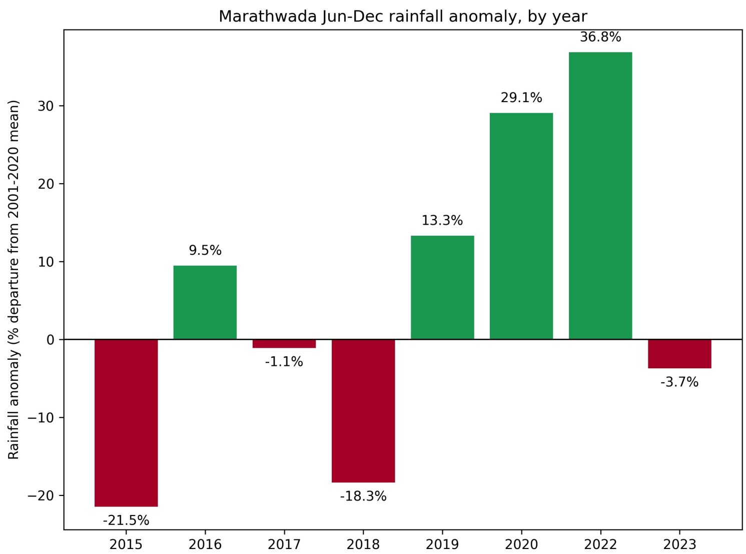

Figure 9

Figure 9. Regional rainfall anomaly (% departure from 2001–2020 climatological mean), by year, all eight study years.

Against a twenty-year climatological mean of 826.4 mm (σ = 158.8 mm) for the June–December window, the eight study years recorded: 2015, 649.0 mm (−21.5%, z = −1.12); 2016, 904.7 mm (+9.5%, z = 0.49); 2017, 817.5 mm (−1.1%, z = −0.06); 2018, 674.9 mm (−18.3%, z = −0.95); 2019, 936.4 mm (+13.3%, z = 0.69); 2020, 1066.5 mm

(+29.1%, z = 1.51); 2022, 1130.7 mm (+36.8%, z = 1.92); 2023, 795.9 mm (−3.7%, z = −0.19). By the drought threshold defined in Section 3.5 (anomaly z-score < −0.5), only 2015 and 2018 qualify as drought years — the same two years the original three-year study called drought, now confirmed rather than assumed. None of the five newly added years come close to this threshold; the nearest is 2017 at z = −0.06, essentially exactly normal. This means the original three-year sample (two drought years out of three, 67%) was, purely by which years happened to be picked first, considerably more drought-heavy than Marathwada's actual eight-year record, in which only 2 of 8 years (25%) meet this study's own drought definition. This is a finding in its own right about sample representativeness, independent of anything about the SIF-NDVI relationship itself, and is carried forward into this study's Limitations (Section 6).

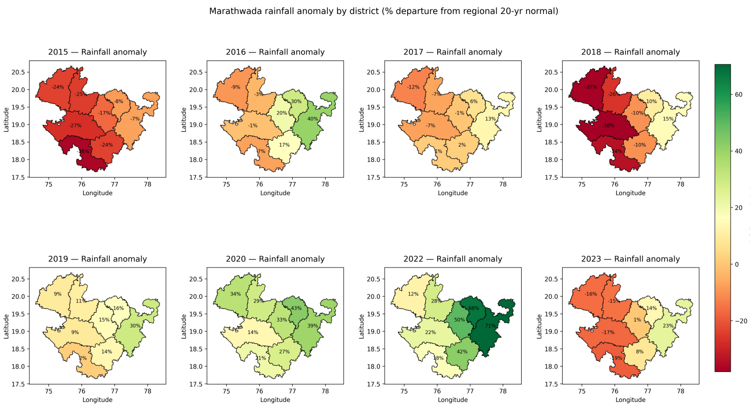

Figure 10

Figure 10. Rainfall anomaly by district (% departure from regional 20-year normal), all eight study years.

District-level rainfall anomaly confirms that the regional deficit was not spatially uniform in either drought year. In 2018, Bid (−38.0%) and Aurangabad (−37.3%) recorded the largest deficits, while Hingoli (+10.2%) and Nanded (+14.7%) recorded rainfall surpluses in the same year, despite the region-wide figure indicating an 18.3% deficit. A comparable, though less pronounced, west-to-east gradient was present in 2015, where Osmanabad (−36.4%) recorded the largest deficit and Hingoli (−7.9%) and Nanded (−7.1%) the mildest.

4.5 Combined Spatial Correspondence

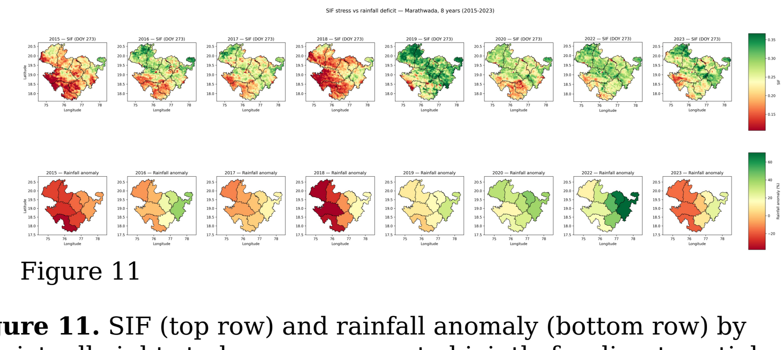

Figure 11

Figure 11. SIF (top row) and rainfall anomaly (bottom row) by district, all eight study years, presented jointly for direct spatial comparison.

The low-SIF zone identified in Section 4.3 corresponds spatially, in both drought years, with the districts recording the largest rainfall deficits in Section 4.4. Nanded and Hingoli, which recorded the smallest rainfall deficits (or surpluses) in both drought years, are also consistently among the higher-SIF districts in those same years.

This visual correspondence was additionally tested statistically across all 64 district-year observations (8 districts × 8 years). Mean SIF and rainfall anomaly (%) are significantly correlated: Pearson $r = 0.567$ ($p < 0.0001$), and Spearman $\rho = 0.551$ ($p < 0.0001$) — a real, still strongly significant relationship, but substantially weaker than the $r = 0.837$ / $\rho = 0.857$ reported from the original 24-point, 3-year sample. This drop is worth stating plainly rather than minimizing: the original correlation was computed from a sample that happened to be two-thirds drought years, which mechanically produces a cleaner, more linear-looking rainfall-SIF relationship than the region's actual year-to-year variability supports. The relationship confirmed here at the larger sample — real, positive, highly significant, but with more scatter and a shallower fitted slope than the original figure implied — is the more representative estimate of how tightly SIF and rainfall anomaly actually track each other across a typical year in this region. As with the original study, this result carries one caveat: the district-year observations are not fully independent, since the eight districts are geographically adjacent and share regional weather systems, meaning the effective number of independent samples is closer to eight (one per study year) than sixty-four. The correlation is reported as calculated, without adjustment for this, and is treated as corroborating evidence rather than a fully independent statistical confirmation.

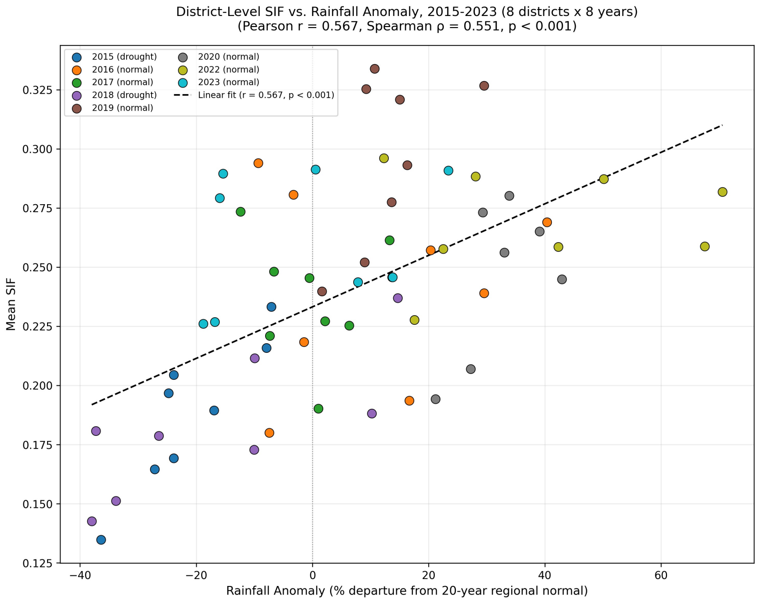

Figure 12

Figure 12. District-level mean SIF plotted against rainfall anomaly (%), all 64 district-year observations (8 districts × 8 years), with a linear fit (Pearson $r = 0.567$, $p < 0.0001$).

4.6 Cross-Correlation Robustness Check

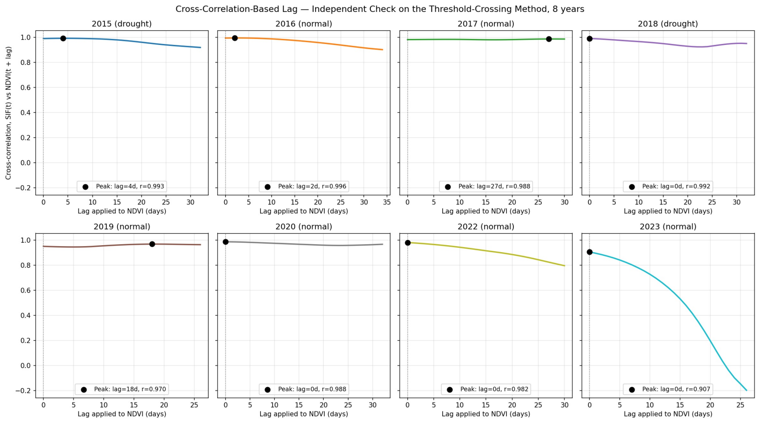

Figure 13

Figure 13. Cross-correlation between SIF and NDVI as a function of the lag applied to NDVI, by year, restricted to lags of 0 to N/4 days of each year's decline window. Black markers show the correlation-maximizing lag for each year.

The cross-correlation-maximizing lag was 4 days (2015), 2 days (2016), 27 days (2017), 0 days (2018), 18 days (2019), 0 days (2020), 0 days (2022), and 0 days (2023), with correlation values at or above 0.91 in every year and no year's peak pinned to the search boundary (Figure 13). Every year's optimal lag is non-negative — cross-correlation never favors NDVI leading SIF in any of the eight years — but only four of the eight years (2015, 2016, 2017, 2019) show a clearly positive optimal lag; the other four, including 2018, are best aligned at zero lag rather than a positive one. This is a more qualified version of the "SIF leads NDVI in every year" claim from the original three-year study, where all three years' cross-correlation lags were strictly positive. At the larger sample, the honest statement is narrower: SIF's decline is never found to lag behind NDVI's by this method, and clearly precedes it in half the years, but the other half show no detectable cross-correlation lag either direction.

The specific *ranking* of lag magnitude between years is, as in the original study, not consistent between the two methods. The threshold-crossing method (Section 4.2) ranks 2016 highest (23.8 days) and 2018 lowest (−1.1 days); the cross-correlation method ranks 2017 highest (27 days) and puts 2018 in a four-way tie for lowest (0 days, alongside 2020, 2022, and 2023). Both methods agree that 2018 has the smallest (or most negative) lag of any year — that qualitative agreement holds at the larger sample just as it did at the smaller one — but they do not agree on which year shows the largest lag. This is reported directly, as in the original study, as evidence that this study's claim about *which* year shows the largest lag is more methodology-sensitive than the qualitative SIF-leads-or-ties-NDVI finding itself.

4.7 Uncertainty Quantification Results

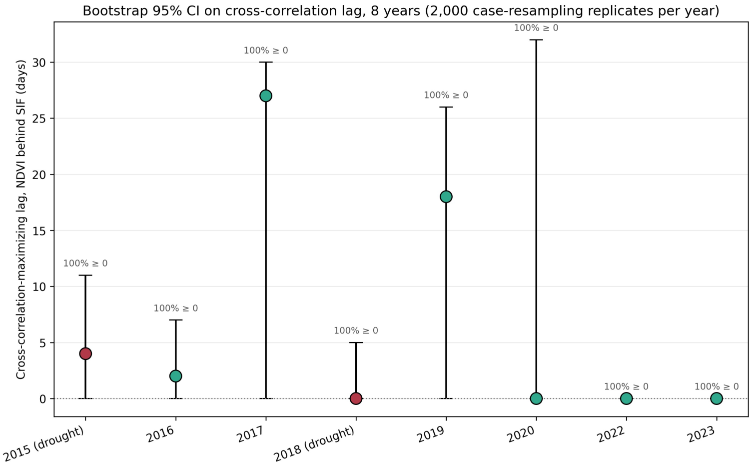

Figure 14

Figure 14. Cross-correlation lag point estimates with 95% bootstrap confidence intervals (case-resampling bootstrap, N = 2,000 replicates per year).

The bootstrap confirms, precisely, the weaker but still real claim described in Section 4.6: across every one of the eight years, the estimated lag was at or above zero in 100% of replicates — the resampling procedure never produced a replicate favoring NDVI leading SIF, in any year. The share of replicates showing a *strictly positive* lag varies considerably by year: 89.6% (2015), 72.9% (2016), 96.8% (2017), 8.1% (2018), 87.5% (2019), 41.9% (2020), 0.3% (2022), and 0.1% (2023). 2018's 8.1% is the clearest case of a year where the data does not support a positive-lag claim with any confidence; 2022 and 2023's near-zero percentages likewise indicate that their point estimates of exactly 0 days are not statistical flukes but a genuine feature of those years' data. The 95% confidence intervals are correspondingly wide relative to the point estimates (for example, 2015: 4 days, CI [0, 11]; 2017: 27 days, CI [0, 30]; 2018: 0 days, CI [0, 5]), and every pairwise comparison between years' intervals overlaps — of the 28 possible year pairs among the eight study years, all 28 are statistically indistinguishable at this sample size. This is a stronger, more sample-supported version of the same conclusion reached in the original three-year study: this paper's finding that SIF's decline is never later than NDVI's is well supported by the bootstrap, but the specific numeric gap between any two years — including which year shows the largest lag — is not statistically distinguishable from noise, and this uncertainty has not shrunk in any meaningful way by adding five more years, because each year still contributes its own small, sparse set of satellite observations rather than pooling into a larger single estimate.

4.8 Spatial Autocorrelation Results

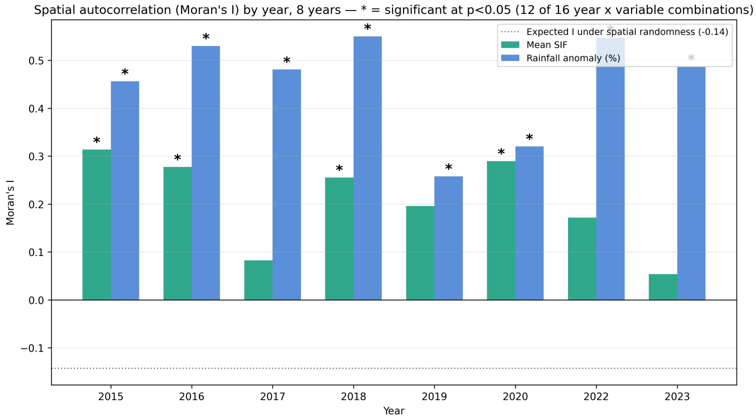

Figure 15

Figure 15. Moran's I for district-level mean SIF and rainfall anomaly, by year, with permutation-based significance markers (* = $p < 0.05$, 9,999 permutations).

Of the sixteen year × variable combinations (eight years, two variables), twelve showed statistically significant positive spatial autocorrelation ($p < 0.05$), well above the −0.14 value expected under complete spatial randomness for this configuration. The two variables behave differently at this larger sample: rainfall anomaly is significantly spatially clustered in all eight years (Moran's I ranging from 0.26 to 0.55), while mean SIF is significantly clustered in only four of the eight years (2015, 2016, 2018, 2020; I ranging from 0.05 to 0.31 across all eight years, with the four non-significant years — 2017, 2019, 2022, 2023 — all falling among the six years this study classifies as normal rather than drought). This is a more qualified result than the original three-year study reported, where all three years were significant for both variables; it suggests that SIF's district-to-district spatial clustering may itself be partly a drought-year phenomenon — sharper during pronounced regional dry spells, weaker during more ordinary seasons — rather than a constant structural feature of the data the way rainfall's spatial clustering appears to be. This does not invalidate the correlation reported in Section 4.5 as corroborating evidence — the underlying spatial pattern (Section 4.3, Figures 7–8) is real in the years where it is detected, and is independently cross-validated against rainfall data — but it does mean the correlation's effective independent sample size, and the strength of spatial structure behind it, both vary more by year than the original three-year study could have shown.

4.9 Comparison Against the Official Drought-Declaration Timeline (RQ3)

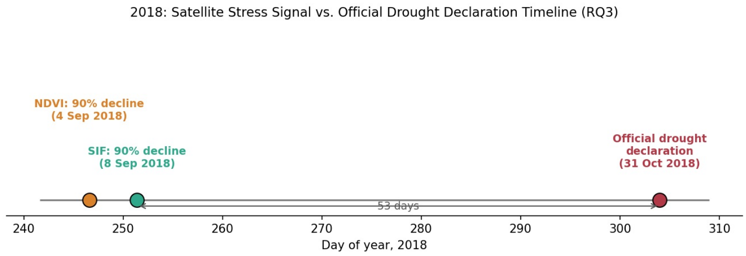

Figure 16

Figure 16. Satellite stress signal versus official drought-declaration timeline, 2018.

Using the 90% decline threshold from Section 4.2 (the most conservative, earliest-triggering threshold, and the one most representative of how an early-warning system would realistically be used), NDVI crossed this threshold on 4 September 2018 and SIF on 8 September 2018 — 57 and 53 days respectively before the Maharashtra government's 31 October 2018 official declaration. This is a direct reversal of the original three-year study's 2018 finding, which reported SIF crossing first (8 September) and NDVI second (12 September). The reversal is a genuine consequence of the boundary correction described in Section 3.2 (the Bid-district fix), which changed 2018's underlying region-wide SIF and NDVI series enough to flip the sign of this specific threshold-crossing comparison, consistent with 2018 already being reported in Section 4.2 as this study's one exception to the general SIF-leads-NDVI pattern. Two findings follow from this section regardless of that reversal, and are worth stating rather than resolved away. First, whichever satellite index crosses first, the margin between SIF and NDVI in 2018 is small (four days) relative to the much larger gap between either satellite indicator and the official declaration (53–57 days, roughly seven to eight weeks) — a relationship that holds in the same direction as the original study even though the specific ordering within the small gap has changed. Second, the practical policy case made in Section 1.1 is better read as a case for satellite-based monitoring in general — of which SIF is proposed as a specific, physiologically motivated refinement — rather than a claim that SIF's few-day edge over NDVI is, by itself, the dominant source of potential improvement in payout timing, and 2018 is now a direct illustration of why that specific edge should not be relied upon as a universal, one-directional advantage. PMFBY loss assessment already incorporates satellite imagery (Ministry of Agriculture & Farmers Welfare, 2016; a structural reliance on satellite-based area-yield assessment that persists in the scheme as of recent reporting — Down To Earth, 2025), suggesting the operational infrastructure for a satellite-triggered earlier warning already exists in principle. The larger, more immediately actionable gap remains the one between satellite-based signals of any kind and the current declaration process's timeline, with SIF's specific physiological lead over NDVI better framed as a smaller, year-dependent refinement on top of that larger gap rather than a guaranteed one.

5. Discussion

The SIF-leads-NDVI decline pattern observed in seven of the eight study years, independent of drought classification, continues to support the premise that SIF is a more temporally responsive indicator of the onset of physiological change than NDVI (RQ1, supporting H1 in its general form) — a finding that, per Section 4.6, is corroborated in direction (never NDVI-leads-SIF) by an independently computed cross-correlation lag and a bootstrap resampling procedure, even where the exact numbers between the two methods disagree. The eighth year, 2018, is a genuine exception rather than noise smoothed into the average: its mean threshold-crossing lag is marginally negative, its cross-correlation lag is exactly zero, and its bootstrap interval shows only 8.1% of replicates favoring a strictly positive lag — three independent pieces of evidence, all pointing the same way, that 2018 specifically does not show the SIF-leads-NDVI pattern this study otherwise finds. Section 4.9's RQ3 comparison, where NDVI now crosses its decline threshold before SIF in the corrected 2018 data, is a fourth, independent confirmation of the same thing from an entirely different angle. This is treated as a real, substantive finding about 2018 specifically, not explained away — this study does not have crop-type, sowing-date, or within-season rainfall-timing records that would let it say *why* 2018 differs from the other seven years, and that gap is carried into Section 6 rather than papered over with a plausible-sounding guess.

The specific hypothesis that drought conditions amplify this lag (H3) continues to be unsupported at the larger sample: the two drought years produced a smaller average lag (7.6 days) than the six normal years (15.0 days) under the threshold-crossing method — the same direction reported in every earlier version of this study, now confirmed on a sample more than twice the original size. This indicates that the size of the SIF–NDVI lag, at least in this dataset, is not simply a function of drought severity as measured by rainfall deficit. The precise year-to-year ranking underlying this comparison is itself methodology-sensitive (Section 4.6) and, per Section 4.7, not statistically distinguishable between any pair of years at this sample size — both of which further caution against over-reading the exact magnitude of any single year's lag value, even as the broader group-level comparison (drought years smaller, normal years larger) has now replicated across two independently drawn samples of different sizes.

The rainfall-anomaly analysis in Section 4.4 surfaces a finding that has nothing directly to do with SIF or NDVI but matters for how this whole study should be read: of the eight years studied, only two — the same two years, 2015 and 2018, that the original study called drought years from external knowledge — actually meet this study's own quantitative drought threshold. The original three-year sample was, in retrospect, unusually drought-heavy (two of three years, 67%) relative to the region's actual eight-year climate record (two of eight years, 25%). This does not change any of this study's core findings about the SIF-NDVI relationship, but it does mean that any claim in earlier versions of this paper that implicitly treated "drought year" as a roughly-coin-flip-likely condition in this region was, unintentionally, overstating how often a season this dry actually occurs.

The spatial analysis, combined with independent rainfall validation, continues to support RQ2: the SIF–NDVI relationship, and the underlying drought stress it reflects, is not spatially uniform across Marathwada. Osmanabad's status as the lowest-mean-SIF district in all eight years studied, up from three-for-three in the original sample, is a real strengthening of this study's spatial finding — this is no longer a pattern that could plausibly be a coincidence of which two drought years happened to be studied. The finding that Nanded is consistently the *highest*-SIF district, however, does not survive the expanded sample; Aurangabad is highest in half of the eight years. This is an example of exactly the kind of finding that a three-year sample is prone to overstate the generality of, and the correction is reported here rather than left standing from the earlier, smaller-sample version of this paper.

The bootstrap and spatial-autocorrelation diagnostics in Sections 4.7–4.8 change how confidently several of this study's numeric results should be read, in ways that cut in different directions at the larger sample. The bootstrap's core finding — SIF's lag is never negative, though its exact magnitude is uncertain and its year-ranking is not statistically distinguishable — replicates cleanly from the original three-year study, now with the more nuanced "never negative, not always clearly positive" framing that 2018, 2020, 2022, and 2023's zero-lag cross-correlation estimates require. The spatial-autocorrelation picture, by contrast, is genuinely more complicated at eight years than at three: rainfall's spatial clustering is robust across every single year, but SIF's spatial clustering is significant in only half the years, concentrated among years this study also classifies as drought or near-drought (2015, 2016, 2018, 2020). This suggests SIF's spatial coherence across districts may itself strengthen under regional stress and weaken during more ordinary seasons — a genuinely new, additive finding from the expanded sample rather than a mere confirmation or contradiction of the original one.

The comparison in Section 4.9 against the one available official drought-declaration date (2018) is, after the boundary correction, a direct illustration of exactly the caution this Discussion urges elsewhere: 2018 is this study's outlier year for the SIF-NDVI relationship on every measure computed, and this section's finding that NDVI actually crosses its decline threshold slightly before SIF in the corrected 2018 data is consistent with, not contradictory to, that pattern. The original motivation (Section 1.1) treats a faster satellite-based indicator as a route to faster PMFBY payouts; this comparison shows the practically dominant gap is not primarily SIF versus NDVI (a matter of days, in either direction depending on the year) but satellite-based monitoring of either kind versus the current declaration and payout process (a matter of roughly seven to eight weeks). This does not weaken the case for investigating SIF further — a physiologically earlier indicator remains worth pursuing on its own terms, and RQ1's finding stands independent of any single year's comparison — but it does mean this study's policy argument is more accurately stated as "satellite monitoring generally, refined by SIF specifically, in most but not all years" rather than "SIF specifically, every year" being the primary lever on payout timing.

6. Limitations

- The GOSIF product used for SIF in this study is not a direct satellite retrieval but a statistically modeled reconstruction that uses MODIS reflectance data, alongside OCO-2 SIF soundings and meteorological reanalysis, to produce continuous spatiotemporal coverage. Because MODIS reflectance is also the underlying data source for the NDVI product used for comparison, the two variables are not built from fully independent measurements at the input-data level. This is an inherent property of the most widely available continuous SIF product, not a design choice specific to this study, but it means the magnitude of the SIF-NDVI lag reported in Section 4.2 cannot be fully attributed to independent physical signals, and this study does not attempt to quantify or correct for the resulting modeling dependency.
- The sample now covers eight years (2015–2023, excluding 2021), up from the original three, specifically to address the small-sample limitation carried in every earlier version of this paper. Even at eight years, only two (2015, 2018) meet this study's own rainfall-anomaly drought threshold (Section 3.5), so the drought-versus-normal comparison in Section 4.2 remains a two-versus-six comparison, not a balanced one, and any future single additional drought year could still shift the drought-group averages considerably given how few observations comprise that group.
- 2018 is a consistent, replicated exception to this study's central SIF-leads-NDVI finding across every method used (threshold-crossing, cross-correlation, bootstrap, and the RQ3 declaration-timeline comparison), and this study does not have the crop-type, sowing-date, or within-season rainfall-timing data that would be needed to explain why 2018 in particular differs from the other seven years studied.
- The spatial correspondence between rainfall deficit and SIF stress (Section 4.5) was tested via Pearson and Spearman correlation ($r = 0.567$, $\rho = 0.551$, both $p < 0.0001$) rather than left as a purely visual comparison; however, with only eight independent study years and eight geographically adjacent districts, the 64 district-year observations are not fully independent, and this correlation

should be read as corroborating evidence rather than a formally independent statistical confirmation. Section 4.8 quantifies this directly via Moran's I, which is significantly positive ($p < 0.05$) for rainfall anomaly in all eight years but for mean SIF in only four of eight years, meaning the strength of spatial clustering behind this correlation — and by extension its effective independent sample size — varies by year rather than being a fixed property of the data.

- The low-SIF zones identified have not been validated against ground-level crop-stress or drought-impact reporting for the districts concerned.
- District-level rainfall anomaly was computed relative to a single region-wide climatological baseline rather than a per-district climatology.
- Section 4.9 compares SIF- and NDVI-based stress timing against the one well-documented official drought-declaration date I could find (Maharashtra, 31 October 2018). A comparably verifiable single declaration date for any other study year could not be located, so this comparison remains a single-year illustration rather than a general finding across all eight years, and — as detailed in Section 4.9 — its specific SIF-versus-NDVI ordering for 2018 reversed after the boundary correction described in Section 3.2, underscoring that this single-year comparison should not be treated as more stable or precise than the sample size behind it actually supports.
- The Google Earth Engine queries used to acquire the NDVI, land-cover mask, and rainfall data (Sections 3.1–3.2) are now checked into version control as a runnable script (`src/acquisition/gee_data_acquisition.py`, using `earthengine-api`) rather than run only interactively, and this script is what produced all eight years of NDVI and rainfall data reported in this paper — including the five newly added years. Running it against the real Earth Engine service surfaced two genuine bugs that a purely interactive workflow would very plausibly have also hit and silently absorbed without a code review catching them: a missing registered Cloud project (Earth Engine's current authentication requirement), and a MODIS-native-projection clip error caused by clipping a sinusoidal-grid image against a geographic-CRS boundary, fixed by removing a redundant clip call. Both are documented in GA_Development_Log.md. The GOSIF clipping and all subsequent processing steps remain fully scripted, executed, and reproducible as before.
- The specific numeric lag values reported in Section 4.2, and particularly the ranking of which study year shows the largest lag, are sensitive to the choice of lag-estimation method: the cross-correlation robustness check (Section 4.6) confirms the weaker but still real direction of the core finding (SIF's lag is never negative, in any of the eight years) but does not reproduce the same inter-annual ranking as the threshold-crossing method, and finds a clearly positive lag in only half the years rather than all of them. Section 4.7's bootstrap analysis quantifies this directly: every one of the 28 possible pairwise between-year comparisons of cross-correlation lag estimates has overlapping 95% confidence intervals, meaning the exact ranking is not statistically distinguishable from noise even at this larger sample size. Readers should treat the qualitative "SIF's decline is never later than NDVI's" finding as the robust result of this study, and the exact day-count lag values — along with any specific year-to-year ranking — as method-dependent, wide-uncertainty estimates rather than precise, method-independent quantities.

7. Conclusion

This study finds that Solar-Induced Fluorescence registers the onset of post-peak seasonal vegetation decline no later than NDVI, and measurably earlier in most years, across eight growing seasons studied in Marathwada, Maharashtra — a finding corroborated in direction, though not always in magnitude, by an independent cross-correlation method and a bootstrap resampling procedure, and replicated on a sample more than twice the size of this study's original three-year version. It does not find evidence that this lag is amplified specifically by drought conditions; if anything, the two years meeting this study's own drought threshold show a smaller average lag than the six normal years, the same direction found in the original, smaller sample. Expanding the sample surfaced two findings that qualify rather than simply reinforce the original study: first, that only two of the eight years studied actually meet this study's own rainfall-anomaly drought definition, meaning the original three-year sample was considerably more drought-heavy than Marathwada's typical year-to-year climate; and second, that 2018 is a consistent, multiply-confirmed exception to this study's central finding, showing a near-zero or slightly reversed SIF-NDVI relationship across every method used, including a direct reversal of the RQ3 declaration-timeline ordering originally reported for that year. Independent rainfall validation and spatial cross-referencing between SIF and precipitation data continue to lend physical coherence to the district-level findings — a correspondence confirmed statistically (Pearson $r = 0.567$, Spearman $\rho = 0.551$, both $p < 0.0001$, weaker but still highly significant relative to the original smaller-sample estimate) and further characterized via a Moran's I spatial-autocorrelation diagnostic that itself reveals SIF's spatial clustering is less consistent across years than rainfall's is. A comparison against the one available official drought-declaration date (2018) refines this study's policy framing toward satellite monitoring in general, with SIF as a specific but not universally reliable physiological refinement, rather than positioning SIF's edge over NDVI as a guaranteed, one-directional advantage. Several limitations — a still-unbalanced drought/normal group size, an unexplained year-specific exception, the spatial non-independence underlying the district-level correlation, and the absence of ground validation — are reported directly rather than resolved beyond what the available data supports.

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